Types of Operating Systems

Operating Systems can be categorized according to different criteria like whether an operating system is for mobile devices (examples Android and iOS) or desktop (examples Windows and Linux).

Batch Operating System

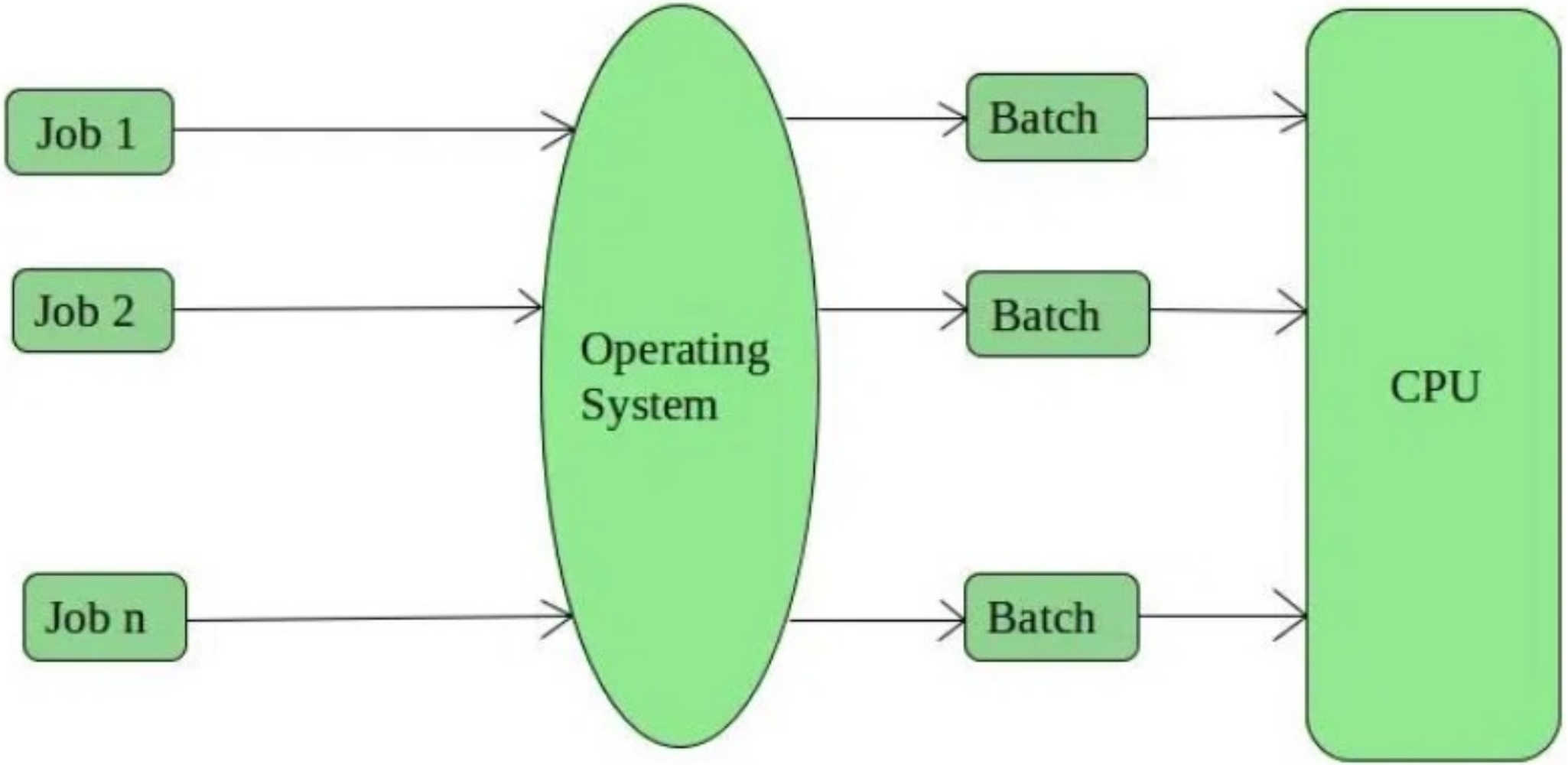
This type of operating system does not interact with the computer directly. There is an operator which takes similar jobs having the same requirements and groups them into batches.

It is the responsibility of the operator to sort jobs with similar needs.

Batch Operating System is designed to manage and execute a large number of jobs efficiently by processing them in groups.

Types of Operating Systems

Batch Operating System



Types of Operating Systems

Advantages of Batch Operating System

- Multiple users can share the batch systems.
- The idle time for the batch system is very less.
- It is easy to manage large work repeatedly in batch systems.

Disadvantages of Batch Operating System

- CPU is not used efficiently. When the current process is doing IO, CPU is free and could be utilized by other processes waiting.
- The other jobs will have to wait for an unknown time if any job fails.
- In batch operating system, average response time increases as all processes are processed one by one.

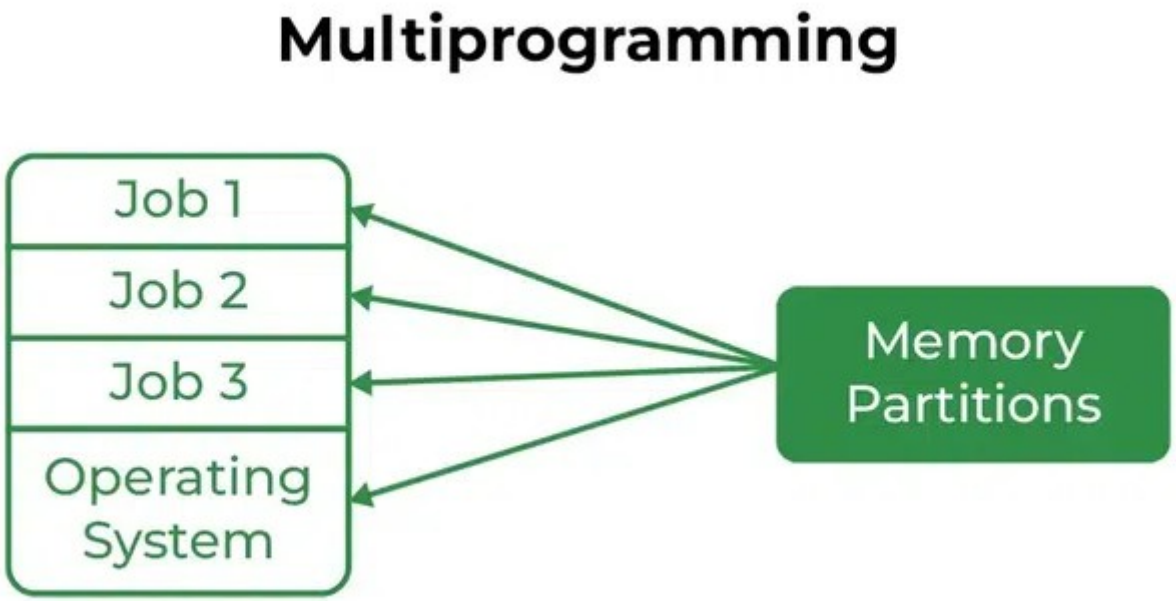
Types of Operating Systems

Multi-Programming Operating System

Multiprogramming Operating Systems can be simply illustrated as more than one program is present in the main memory and any one of them can be kept in execution. This is basically used for better utilization of resources.

Advantages of Multi-Programming Operating System

- CPU is better utilized and overall performance of the system improves.
- It helps in reducing the response time.

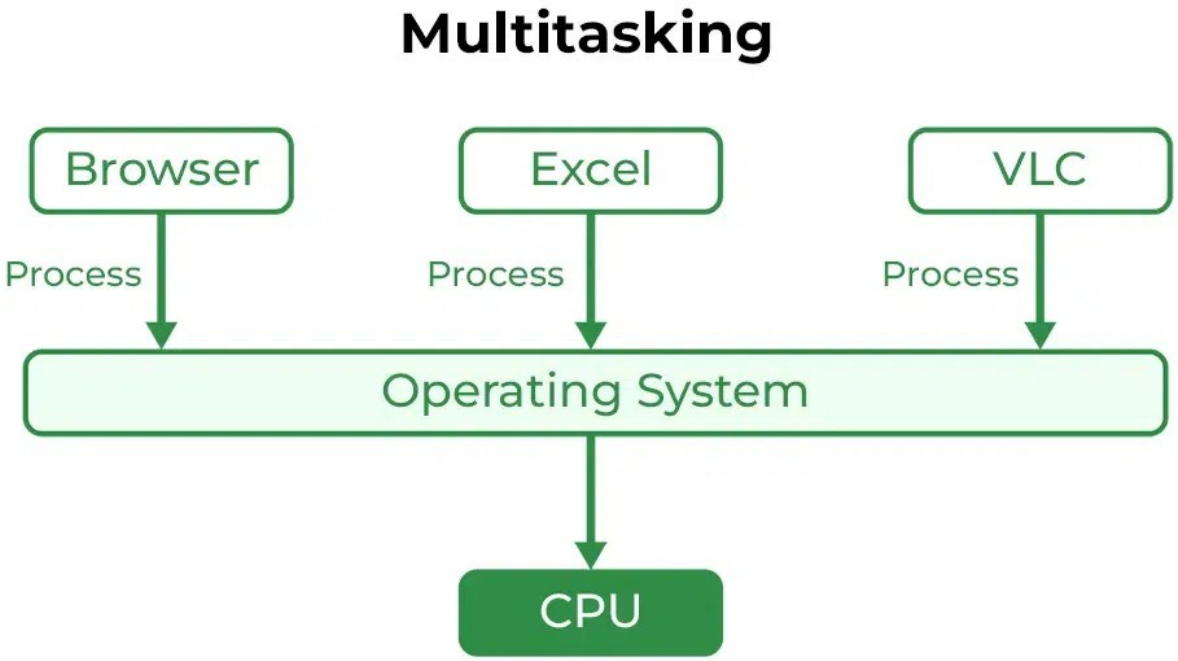


Types of Operating Systems

Multi-Tasking/Time-sharing Operating systems

It is a type of Multiprogramming system with every process running in round robin manner. Each task is given some time to execute so that all the tasks work smoothly. Each user gets the time of the CPU as they use a single system. These systems are also known as Multitasking Systems.

The task can be from a single user or different users also. The time that each task gets to execute is called quantum. After this time interval is over OS switches over to the next task.



Types of Operating Systems

Advantages of Time-Sharing OS

- Each task gets an equal opportunity.
- Fewer chances of duplication of software.
- CPU idle time can be reduced.
- Resource Sharing: Time-sharing systems allow multiple users to share hardware resources such as the CPU, memory, and peripherals, reducing the cost of hardware and increasing efficiency.
- Improved Productivity: Time-sharing allows users to work concurrently, thereby reducing the waiting time for their turn to use the computer. This increased productivity translates to more work getting done in less time.
- Improved User Experience: Time-sharing provides an interactive environment that allows users to communicate with the computer in real time, providing a better user experience than batch processing.

Types of Operating Systems

Disadvantages of Time-Sharing OS

- Reliability problem.
- One must have to take care of the security and integrity of user programs and data.
- Data communication problem.
- High Overhead: Time-sharing systems have a higher overhead than other operating systems due to the need for scheduling, context switching, and other overheads that come with supporting multiple users.
- Complexity: Time-sharing systems are complex and require advanced software to manage multiple users simultaneously. This complexity increases the chance of bugs and errors.
- Security Risks: With multiple users sharing resources, the risk of security breaches increases. Time-sharing systems require careful management of user access, authentication, and authorization to ensure the security of data and

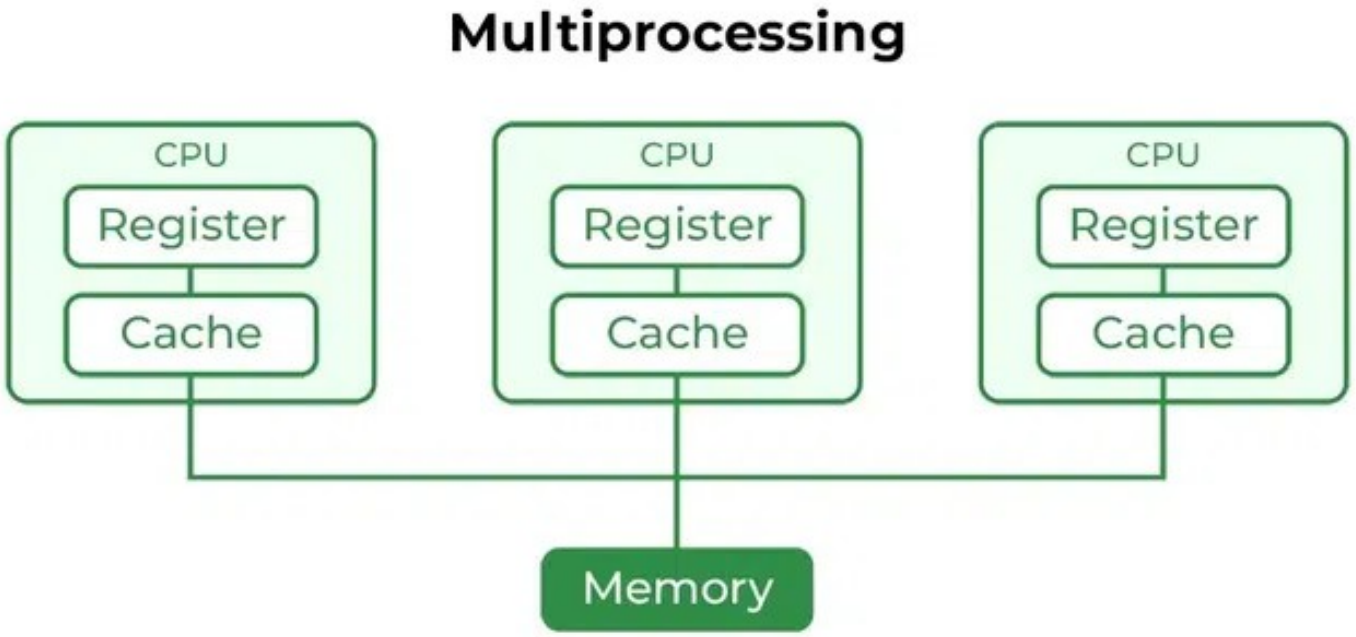
Types of Operating Systems

Multi-Processing Operating System

Multi-Processing Operating System is a type of Operating System in which more than one CPU is used for the execution of resources. It betters the throughput of the System.

Advantages of Multi-Processing Operating System

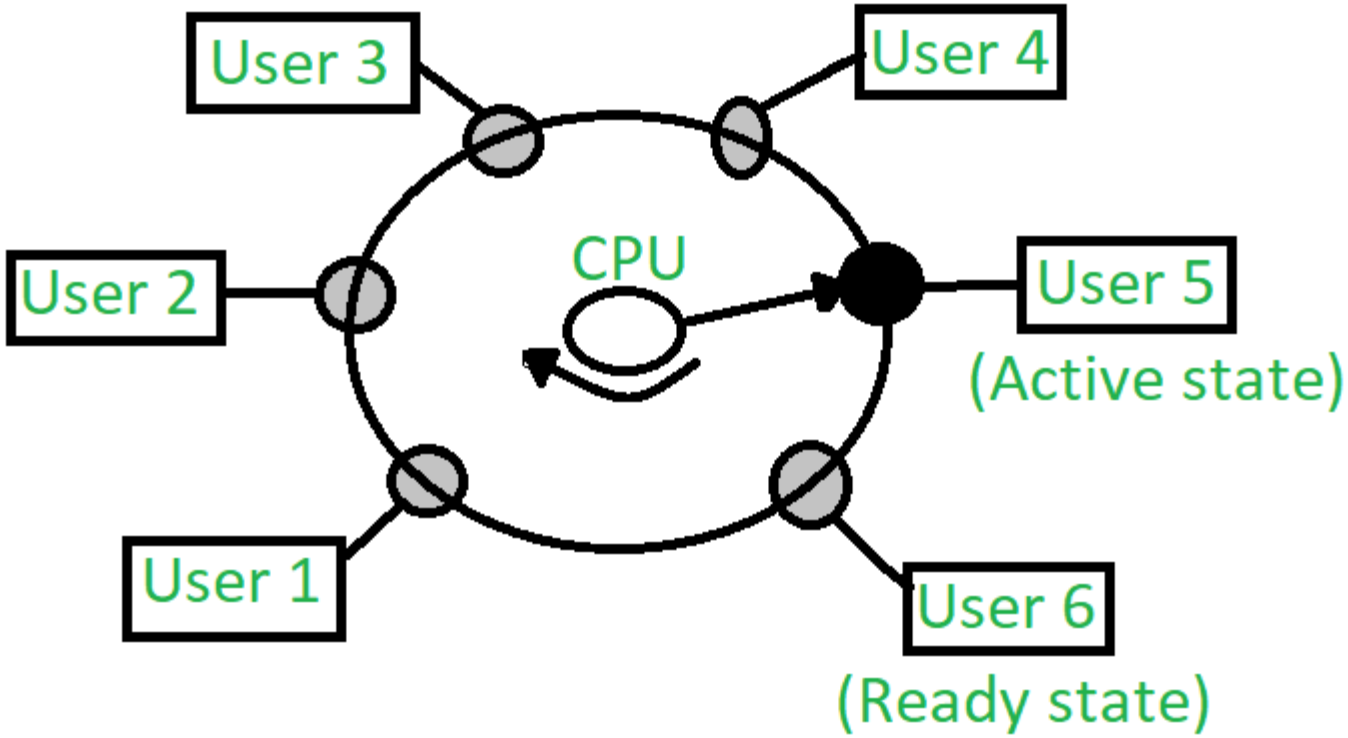
- It increases the throughput of the system as processes can be parallelized.
- As it has several processors, so, if one processor fails, we can proceed with another processor.



Types of Operating Systems

Multi User Operating Systems

These systems allow multiple users to be active at the same time. These system can be either multiprocessor or single processor with interleaving.



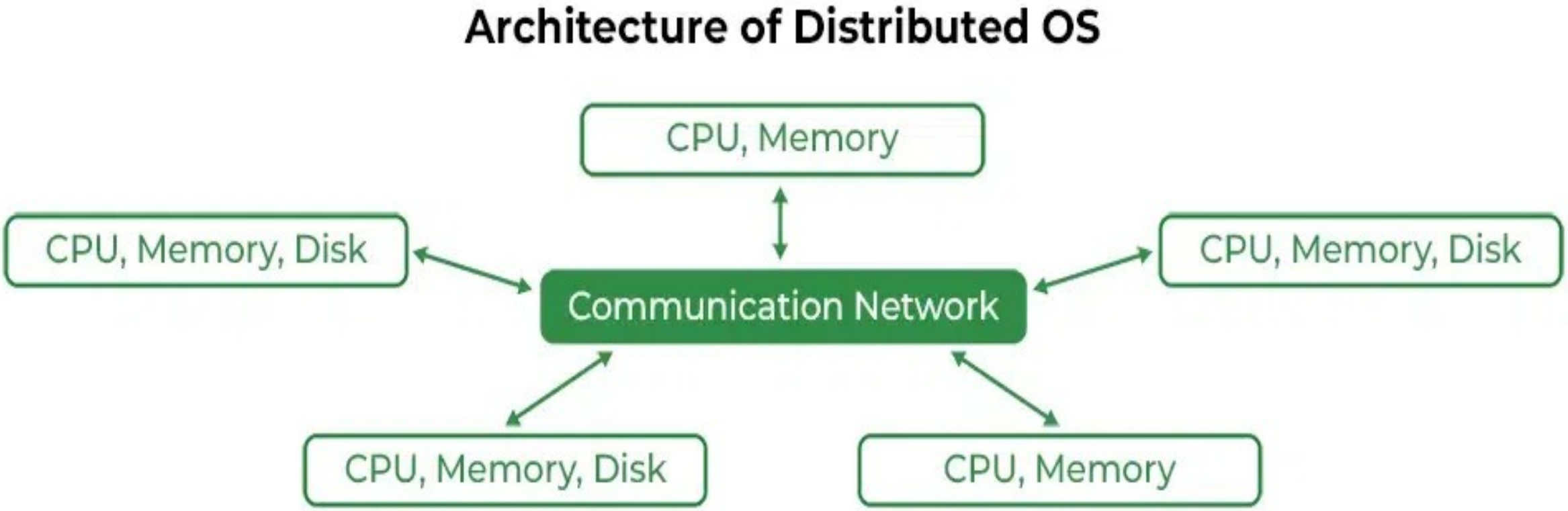
Types of Operating Systems

Distributed Operating System

- These types of operating system is a recent advancement in the world of computer technology and are being widely accepted all over the world and, that too, at a great pace.
- Various autonomous interconnected computers communicate with each other using a shared communication network. Independent systems possess their own memory unit and CPU. These are referred to as loosely coupled systems or distributed systems. These systems' processors differ in size and function.
- The major benefit of working with these types of the operating system is that it is always possible that one user can access the files or software which are not actually present on his system but some other system connected within this network.

Types of Operating Systems

Distributed Operating System



Types of Operating Systems

Advantages of Distributed Operating System

- Failure of one will not affect the other network communication, as all systems are independent of each other.
- Electronic mail increases the data exchange speed.
- Since resources are being shared, computation is highly fast and durable.
- Load on host computer reduces.
- These systems are easily scalable as many systems can be easily added to the network.
- Delay in data processing reduces.

Types of Operating Systems

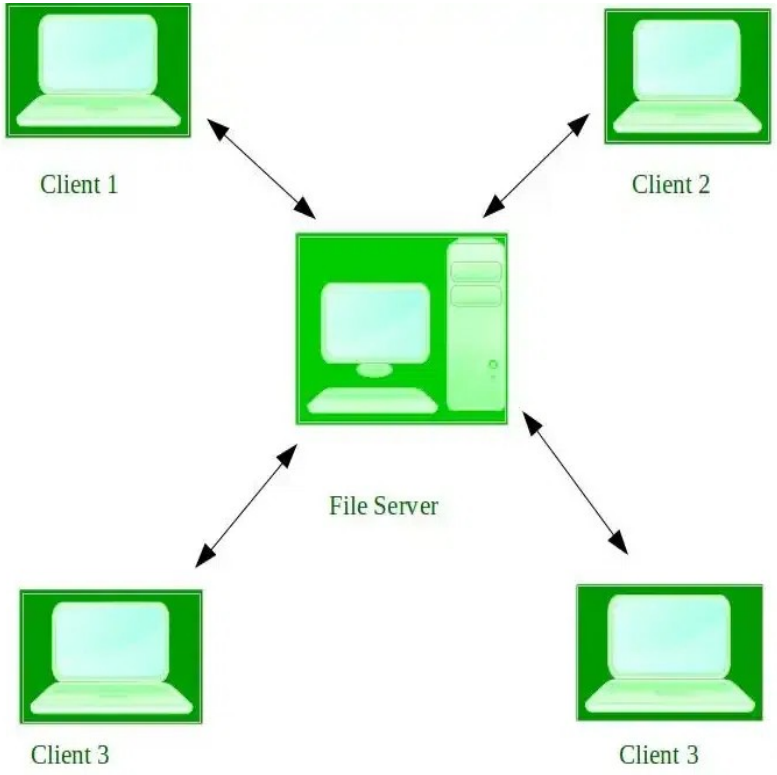
Disadvantages of Distributed Operating System

- Failure of the main network will stop the entire communication.
- To establish distributed systems the language is used not well-defined yet.
- These types of systems are not readily available as they are very expensive.
- Not only that the underlying software is highly complex and not understood well yet.

Types of Operating Systems

Network Operating System

These systems run on a server and provide the capability to manage data, users, groups, security, applications, and other networking functions. These types of operating systems allow shared access to files, printers, security, applications, and other networking functions over a small private network. One more important aspect of Network Operating Systems is that all the users are well aware of the underlying configuration, of all other users within the network, their individual connections.



Types of Operating Systems

Advantages of Network Operating System

- Highly stable centralized servers.
- Security concerns are handled through servers.
- New technologies and hardware up-gradation are easily integrated into the system.
- Server access is possible remotely from different locations and types of systems.

Disadvantages of Network Operating System

- Servers are costly.
- User has to depend on a central location for most operations.
- Maintenance and updates are required regularly.

Types of Operating Systems

Real-Time Operating System

These types of OSs serve real-time systems. The time interval required to process and respond to inputs is very small. This time interval is called **response time. Real-time systems** are used when there are time requirements that are very strict like missile systems, air traffic control systems, robots, etc.

Types of Real-Time Operating Systems

- **Hard Real-Time Systems:** Hard Real-Time OSs are meant for applications where time constraints are very strict and even the shortest possible delay is not acceptable. These systems are built for saving life like automatic parachutes or airbags which are required to be readily available in case of an accident. Virtual memory is rarely found in these systems.
- **Soft Real-Time Systems:** These OSs are for applications where time-constraint is less strict.

Types of Operating Systems

Hard real time system is a system whose operation is incorrect whose result is not produced according to time constraint. In hard real-time systems, timing constraints are extremely strict and failure to meet a deadline can have serious consequences, such as human injury or equipment damage. These systems typically require deterministic and predictable behavior, with a guaranteed response time for critical tasks. Examples include medical equipment, military defense systems, and industrial control systems.

Advantages of Hard Real-Time Systems

- **Guaranteed Response Time:** Through this the system will always respond within the stipulated time and hence is predictable.
- **Suitable for Critical Applications:** Applied in applications such as medical devices or aerospace applications where failure cannot be an option.

Types of Operating Systems

Disadvantages of Hard Real-Time Systems

- **Complex Design:** The stringent requirements which are characteristic of these systems make their design and implementation very difficult.
- **High Cost:** As a rule, it calls for the use of special equipment and software, which is costly to adopt.

Examples of Hard Real-Time Systems

1. Medical Equipment (e.g., Pacemakers, ICU Monitors)

A pacemaker regulates a patient's heartbeat. If it fails to send electrical signals on time, the patient could suffer serious health issues or even death.

2. Military Defense Systems (e.g., Missile Defense Systems, Radar Systems)

A missile interception system must detect and neutralize incoming threats in real-time. A delay of even a few milliseconds can result in mission failure and catastrophic damage.

Types of Operating Systems

Examples of Hard Real-Time Systems

3. **Industrial Control Systems (e.g., Nuclear Power Plant Control, Robotics in Manufacturing)**

   A nuclear reactor control system must regulate temperature and pressure precisely. If it fails to take corrective actions within the required time, it may cause a reactor meltdown.

4. **Automotive Systems (e.g., Airbag Deployment, Anti-lock Braking System - ABS)**

   In an accident, an airbag must deploy within milliseconds of impact. A late response makes it ineffective, risking severe injury or death.

Types of Operating Systems

Illustration

Let’s consider an **anti-lock braking system (ABS)** in a car.

Scenario 1: (Correct behavior - Meeting timing constraints)

- The ABS system detects wheel lock-up and responds within **5 milliseconds** to prevent skidding.
- The car safely stops, preventing an accident.

Scenario 2: (Incorrect behavior - Missing timing constraints)

- The ABS system detects wheel lock-up but responds **after 100 milliseconds** due to system delay.
- By the time braking is applied, the car has already skidded, potentially causing a crash.

Thus, hard real-time systems must always meet their timing deadlines to function correctly and safely.

Types of Operating Systems

Soft Real Time System

A **Soft Real-Time System** is a variant of real-time system where timely processing is desirable but late processing do not cause the system to fail or can cause dramatic loss of life and property. However, failing to meet the time expectation, lead to a decline of service delivery or limited quality output. These systems are often employed in application scenarios where even though timeliness is valuable it is not considered a strict requirement, for instance multimedia streaming applications, online games and data that while processing of which even though they may cause disruptions they do not affect the system criticality of an application. In soft real-time systems, it of no essence to be completely accurate with the heaviest emphasis on flexibility.

Types of Operating Systems

Advantages of Soft Real-Time Systems

- **Flexible Timing:** The delay on the system has showed that the interruption of the frame rate does not affect the gameplay very severely.
- **Easier to Implement:** It is slightly less complex than hard real-time systems and as a result easier and cheaper to design and implement.
- **Cost-Effective:** It is ideal for low risk projects where the use of expensive hardware is avoidable.

Disadvantages of Soft Real-Time Systems

- **Inconsistent Performance:** This means that the operation of the system may at times be slow particularly when there is a high traffic of users.
- **Not Suitable for Critical Applications:** Not suitable for safety-aware applications whereby delays can have catastrophic effects such as system failure or loss of human lives.

Types of Operating Systems

Difference Between Hard Real Time and Soft Real Time System

Types of Operating Systems

Difference Between Hard Real Time and Soft Real Time System

Types of Operating Systems

Mobile Operating Systems

Mobile operating systems (Mobile OS) manage mobile gadgets like phones and tablets. These systems run apps. They are not like desktops as mobiles have different needs. An operating system (OS) for mobile devices manages the basic functions. It runs apps, controls memory, and connects to networks. These systems provide an easy-to-use interface.

Key Features of Mobile Operating system:

User Interface (UI): Touch inputs of Graphical User Interface (GUI) provided by mobile OS are optimized. This is where users can use touch gestures, in other words, swiping, tapping, and pinching, to interact with their gadgets.

Multitasking: It helps in running of many apps at the same time but what is more we can quickly switch between them without any hindrance. Such offloading is for applications that are not currently used actively.

Types of Operating Systems

Key Features of Mobile Operating system:

Connectivity: It provides a variety of connections such as cellular, Wi-Fi, Bluetooth, NFC (Near Field Communication) and others to facilitate the communication of the device with other devices and networks.

Application Management: Is a platform that has its own app marketplace or store which the users utilize to browse, install, run and updates the applications exclusively for that platform.

Resource Management: Efficiently allocates hardware resources like the CPU ,RAM , and battery by achieving a balance between performance and battery life.

Types of Operating Systems

Types of Mobile Operating System

The mobile world has seen many operating systems. Some are big and powerful. Others are small and niche. Here are the main types of mobile OS :

Android: Made by Google, Android is the most used mobile OS worldwide. It's an open-source system built on Linux code. Android is made mostly for touchscreen phones and tablets. Being open allows companies to customize Android for their gadgets. That leads to many different Android devices.

Apple Inc. created iOS, the operating system for iPhone, iPad, and iPod Touch. Its smooth interface and tight Apple ecosystem integration are hallmarks. However, Apple's total control over hardware and software limits customization options, unlike Android's open platform.

Types of Operating Systems

Huawei developed HarmonyOS (Hongmeng OS in China): work across diverse devices like smartphones, laptops, smart home gadgets.

KaiOS is a basic mobile OS: It powers basic phones without touchscreens. KaiOS is based on discontinued Firefox OS. KaiOS supports 4G, GPS, and apps like Facebook and WhatsApp. It offers strong capabilities for non-smartphone mobile devices.

Tizen: It is a Linux-based mobile OS developed by Samsung. It was created with Intel and the Tizen Association. Tizen powers various Samsung products.

Smartphones have different operating systems. Android and iOS are the biggest. But others exist for different users. Operating systems cater to diverse needs. They have pros, cons, and target markets.

Types of Operating Systems

Embedded Operating Systems

Embedded systems are specialized computing systems designed to perform dedicated functions or tasks within larger mechanical or electrical systems. Unlike general-purpose computers, embedded systems are typically designed with specific requirements and constraints in mind, including real-time performance, limited power consumption, and specific form factors. They are integral to many modern technologies, providing essential functionality and control in various applications.

Key Characteristics of Embedded Systems in System Design:

Dedicated Functionality: Embedded systems are tailored to perform specific tasks or functions, often with high efficiency and reliability. This contrasts with general-purpose systems that can run a wide range of applications.

Types of Operating Systems

Real-Time Operation: Many embedded systems operate in real-time, meaning they must process inputs and produce outputs within strict timing constraints. This is crucial for applications like automotive control systems, medical devices, and industrial automation.

Resource Constraints: Embedded systems often operate under stringent resource constraints, including limited processing power, memory, and energy consumption. Design choices must optimize these resources to meet the application's requirements.

Integration and Miniaturization: Embedded systems are typically integrated into larger systems, often requiring compact and efficient designs. This necessitates careful consideration of physical form factors and integration with other system components.

Types of Operating Systems

Reliability and Robustness: High reliability and robustness are critical, especially in applications where failure can lead to significant consequences, such as in aerospace, healthcare, and automotive systems.

Specialized Hardware and Software: The hardware and software in embedded systems are often highly specialized, tailored to the specific needs of the application. This can include custom microcontrollers, real-time operating systems (RTOS), and application-specific integrated circuits (ASICs).

Types of Operating Systems

Advantages of Embedded System

1.High Efficiency and Performance:

Optimization: Embedded systems are specifically designed to perform dedicated tasks, allowing for high levels of optimization in both software and hardware. This leads to increased efficiency and performance compared to general-purpose systems.

Real-Time Processing: Many embedded systems are designed to handle real-time tasks, ensuring that they can process inputs and produce outputs within strict timing constraints.

Types of Operating Systems

Advantages of Embedded System

2.Low Power Consumption:

Energy Efficiency: Embedded systems are often designed with power efficiency in mind, making them ideal for battery-powered devices and applications where energy consumption is a critical concern.

Longer Battery Life: In portable devices, optimized power usage results in longer battery life, which is crucial for user satisfaction and device usability.

Types of Operating Systems

Advantages of Embedded System

3.Compact Size:

Miniaturization: Embedded systems can be designed to be very small, making them suitable for applications with strict size constraints, such as wearable technology, medical implants, and compact consumer electronics.

Space Efficiency: The compact nature of embedded systems allows for more efficient use of space in larger systems, enabling the development of more sophisticated and multi-functional devices.

Types of Operating Systems

Advantages of Embedded System

4.Cost-Effectiveness:

Reduced Cost: Because embedded systems are designed to perform specific tasks, they can be produced with fewer components and resources, reducing manufacturing costs.

Mass Production: The ability to mass-produce embedded systems for specific applications further drives down costs, making advanced technology more accessible.

5. Reliability and Stability:

Dedicated Functionality: Embedded systems are less likely to encounter the software conflicts and errors common in general-purpose systems, leading to higher reliability and stability.

Long-Term Operation: These systems are often designed for long-term, continuous operation without the need for frequent maintenance or updates, making them suitable for critical applications.